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MILITARY SPECIFICATION

TRUCK, WITH HIGH LIFT DUMP MECHANISM: DUMP, CARGO AND STAKE BODY TYPES, DIESEL AND GASOLINE ENGINE DRIVEN, 24,000 GVW, 4x2, COMMERCIAL

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers diesel and gasoline engine driven, four-wheel, two-rear-wheel drive, commercial trucks with a high lift dump mechanism. The truck has a minimum gross vehicle weight (GVW) of 24,000 pounds for normal operating conditions. Vehicles procured under this specification are commercial items which shall be warranted by the manufacturer as specified in procurement documents.

* 1.2 Classification. Vehicles shall be of the types shown in table I, as specified (see 6.2).

TABLE I. Vehicle types.

Type	Nomenclature
I	Coal dump body
II	Cargo dump body
III	Stake dump body

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GBS.1, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Starting.
ZZ-T-381	- Tires, Pneumatic, Vehicular (Highway).

MILITARY

MIL-H-46170	- Hydraulic Fluid, Rust Inhibited, Fire-Resistant Synthetic Hydrocarbon Base.

STANDARDS

MILITARY

MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Rustproofing, Identification Marking and Data Plate Standards.
MS51118	- Pintle Assembly, Towing: 40,000 Lbs. Capacity, Manual Release.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment – Medium and Heavy Trucks.

(Application for copies should reference the Code of Federal Regulations 40 CFR and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

SAE, INC.

SAE Standards and Recommended Practices.

	J245	- Engine Rating Code - Spark Ignition.
	J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DoD adopted).
	J537	- Storage Batteries (DoD adopted).
	J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (20-1000 MHz).
	J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
	J588	- Turn Signal Lamps.
	J589	- Turn Signal Switch.
	J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
	J683	- Tire Chain Clearance - Trucks, Buses and Combinations of Vehicles (DoD adopted).
	J688	- Truck Ability Prediction Procedure (DoD adopted).
	J704	- Openings for Six- and Eight-Bolt Truck Transmission Mounted Power Take-Offs (DoD adopted).
	J816	- Engine Test Code - Spark Ignition and Diesel.
	J844	- Nonmetallic Air Brake System Tubing (DoD adopted).
	J1067	- Seven-Conductor Jacketed Cable for Truck-Trailer Connectors.
*	J1349	- Engine Power Test Code - Spark Ignition and Diesel.

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(Application for copies should be addressed to the SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook.

(Application for copies should be adressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

TRUCK BODY AND EQUIPMENT ASSOCIATION, INC. (TBEA)
Hoist Classification Chart

(Application for copies should be addressed to Truck Body and Equipment Association, Inc., 5530 Wisconsin Avenue, Washington, D.C. 20015).

(Industry association specifications and standards are, generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.l.1.9, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items which meet or exceed the requirements of this specification. All chassis items shall be as represented in the chassis manufacturer's technical data, and special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components and shall be on file in the engineering offices of the procuring activity, prior to delivery of the items.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Painting and marking. As specified by the procuring activity for the appropriate service (see 6.2), treatment, painting, identification marking and data plates shall be in accordance with MIL-STD-1223.

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* 3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with MIL-STD-1223. When specified (see 6.2), tropical rustproofing in accordance with MIL-STD-1223 shall be furnished.

* 3.1.1.3 Wood treatment. Wood surfaces shall be treated in accordance with MIL-STD-1223.

* 3.1.1.4 Drain plugs. Drain plugs installed in the rear axle and manual transmission for the purpose of draining lubricants shall be of the permanent-magnet type.

3.1.1.5 Heavy duty cooling system. When specified (see 6.2), a heavy duty cooling system shall be furnished that shall maintain engine coolant at a temperature below the boiling point with vehicle loaded to rated GVW and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F).

3.1.1.6 Coolant system and indicators. The coolant system shall include a de-aeration system, a surge tank or a coolant recovery reservoir of not less than 2 quart capacity. A high coolant temperature or low coolant level alarm buzzer and red indicator warning light shall be provided on the dash instrument panel.

3.1.1.7 Towing devices. Not less than two hooks, loops or pins for towing the vehicle, shall be furnished on the front of the vehicle.

3.1.1.8 Wheel splash and stone throw protection. The vehicle shall have rigid splash shields ahead of the rear wheels and rubber mud flaps to the rear of the rear wheels. Splash shield and mud flap installations shall conform to the rear wheel splash and stone throw protection provisions of SAE J682.

3.1.1.9 Trailer towing package. When specified (see 6.2), a trailer towing package shall be furnished. The trailer towing package shall consist of a pintle, safety chain attachment devices, a lighting receptacle, a trailer brake control system, and associated reinforcements and wiring, and shall be installed on the rear of the vehicle. The pintle shall be of the rotating type conforming to MS51118 and shall be installed on the chassis frame with reinforcements to transfer pintle loads directly to the chassis rails. The rearmost portion of the pintle shall be forward, but not more than 4 inches forward, of the rearmost part of the vehicle. Two trailer

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safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength at least equal to the GVW of the truck furnished. The safety chain attachment devices shall be capable of accommodating a standard grab hook (4-9/16 inches wide, 1-3/16 inches thick, 25/32 inch throat width) for a 5/8 inch chain. The lighting receptacle, conforming to SAE J560 with its conductors connected and color coded as specified therein, shall be mounted in a readily accessible location near the pintle. The trailer brake control system shall conform to 3.4.11.2.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

* 3.2.2 Air pollution control. The vehicle shall comply with EPA Regulations Governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

* 3.2.3 Sound level. The interior sound level shall conform to Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to EPA Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis and cab, with all attachments, accessories, and equipment; body; and a full complement of fuel, lubricants, and coolant.

3.2.5 Gross vehicle weight. Gross vehicle weight (GVW) shall consist of the curb weight, operator (weight computed at 175 pounds) and a payload to provide not less than the specified GVW.

3.2.6 Weight distribution. The distribution of the GVW for the purpose of establishing suspension, axle and tire capacities shall be determined with the payload uniformly distributed over the load area.

3.2.7 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings must be submitted to the engineering office of the procuring activity. Minimum GVW rating shall be 24,000 pounds.

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* 3.2.8 Overall width. The overall width of the vehicle exclusive of mirror, lights, reflectors and tires shall be not more than 96 inches.

3.3 Performance.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with truck loaded to specified GVW.

3.3.1.1 High speed gradeability. The vehicle shall ascend a continuous 1.0 percent grade at 50 miles per hour (mph). Gradeability requirements shall be met with the main transmission in direct drive, and when multi-speed axle 18 furnished, with the axle in high speed range. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

3.3.1.2 Low speed. Low speed shall be calculated with the engine operating at not less than 35 percent of recommended governed speed, and shall provide not more than the vehicle speed (mph) specified in table II for the corresponding number of available forward speeds.

* TABLE II. Low speed requirement.

Number of forward speeds	Maximum low speed (mph)
4 or 5	5.0
7 or 8	4.0
10	3.0

* 3.3.1.3 Maximum geared speed. Maximum geared speed at engine governed speed shall be not less than 60 mph (minus 1 percent). Conformance to geared speed shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3)}}$$

3.3.2 Service brakes. The service brakes shall control and hold vehicle, when loaded to its specified gross vehicle weight, on a 30 percent grade. The service brakes shall stop the vehicle within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52 under all conditions of loading up to specified GVW.

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3.3.3 Subframe, hoist and body performance.

* 3.3.3.1 Stability. Stability shall be demonstrated with and without payload. The body shall be stable and rigid throughout the entire elevation, retraction and dumping of the hoist mechanism. In the fully elevated position, the body floor shall not deflect more than 1-1/2 inches, measured from the floor, to the top of the subframe. The vehicle in all elevated and dumping positions shall show no evidence of any tire departing from the ground.

* 3.3.3.2 Lifting capacity. The rated lifting capacity of the hoist shall be not less than the sum of the weight of the body plus one and one half times the payload.

* 3.3.3.2.1 Height. The hoist shall be capable of raising the body and payload from rest position to a minimum height of ten feet measured from the ground to the body floor.

* 3.3.3.3 Manual locking device. A manual locking device shall be provided to retain the body and hoist at a raised height of not less than 96 inches measured from ground level to the body floor.

* 3.3.3.4 Operating speeds. The hoist and body shall be capable of elevating and retracting with the payload at not less than 15 feet per minute (fpm) and not more than 20 fpm, with the vehicle engine at idle speed, and when the load is imposed on the hydraulic system. The hydraulic fluid shall be stabilized at the ambient air temperature prior to the performance of the test.

* 3.3.3.5 Position stability. When the vehicle is on level ground, the hoist hydraulic system shall hold the body with the payload in any elevated position without more than a 1/8-inch vertical setting in 2 hours.

* 3.3.3.6 Unbalanced load. With one quarter of the payload located in any corner of the body floor, the hoist shall be capable of raising and lowering the body without evidence of binding, deformation, or bending of the hoist mechanism. The body shall not deflect more than 1-1/2 inches, as specified in 3.3.3.1.

* 3.3.3.7 Wind resistance. The vehicle, both loaded and unloaded, and at the fully elevated position on a 3 percent sidegrade, shall be capable of withstanding 30 mph winds and wind gusts up to 50 mph from any direction without overturning and without permanent deformation of any component. Wind load shall be calculated with the maximum side area perpendicular to the wind force, and shall be measured in accordance with the following:

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(a) at the body, when the body is fully retracted
(b) at the body and hoisting mechanism, when the body is elevated to its maximum specified height, in the most critical position, that would induce tipping.

In both (a) and (b), the resistant moment shall be greater than the wind load. The contractor shall furnish calculations or a certified test report for the vehicle or an identical vehicle to the engineering offices of the procuring activity.

* 3.3.3.8 Stress analysis. A stress analysis of the vehicle body and frame shall be furnished as specified in 3.8.

3.4 Chassis components.

3.4.1 Engine. The engine furnished shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meats or exceeds the requirements of this specification.

* 3.4.1.1 Diesel engine. Unless otherwise specified (see 3.4.1.2), the vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than four cylinders. Engine net horsepower figures used in performance prediction calculations shall be determined in accordance with SAE J816 or SAE J1349. A fan clutch to reduce fan speed automatically, when not required for engine cooling, shall be provided.

3.4.1.2 Gasoline engine. When specified (see 6.2), the vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine, with not less than six cylinders. The engine furnished shall produce the required vehicle performance when operated on unleaded fuel with a research octane rating of 91, at engine speed not more than manufacturer's recommended operating speed. The engine shall be capable of warranted operation on unleaded fuel, when used in accordance with the operator's manual. Unless otherwise specified or required by foreign laws, vehicles for overseas use shall be capable of accepting and operating on leaded gasoline. The rated net torque output of the engine shall be not less than 265 pound-feet. Engine net torque ratings and net horsepower figures used in performance prediction calculations shall be determined in accordance with SAE J245 or SAE J1349. A fan clutch to reduce fan speed automatically, when not required for engine cooling, shall be provided.

3.4.1.3 Oil filter. A full flow type oil filter shall be furnished.

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3.4.1.4 Governor. An engine governor shall be furnished and set and sealed to limit the engine to the engine manufacturer's maximum recommended operating speed.

3.4.1.5 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. On diesel engine driven vehicles, the control shall include partial thermostatic control of coolant flow through the radiator and thermostatically controlled radiator shutters or complete thermostatic control of all coolant flow through the radiator.

* 3.4.1.6 Silicone rubber hoses. When specified (see 6.2), silicone rubber radiator and heater hoses shall be furnished.

* 3.4.1.7 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater; an engine oil heater; and a fuel warmer (diesel engine driven vehicles only), shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. Heaters shall operate on 110-volt alternating current (ac), and shall be wired through a junction block to a single three pronged (male), weatherproof, slave receptacle for receiving external power and grounding vehicle. A three wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. Connecting cable shall include a matching female connector at the vehicle end and a standard three pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to minus 60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

* (a) Coolant heater, 1500-watt minimum rating, shall be installed in the engine block or lower, coolant, inlet hose. Engine thermostat with an operating range of 170°F to 195°F shall be installed;
(b) Immersion type engine oil heater, 300-watt minimum rating, with 170°F to 195°F thermostat, shall be installed in oil pan through any convenient opening;
(c) Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperatures as low as minus 60°F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F;

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(d) A fuel warmer or pre-heater shall be provided on diesel engine driven vehicles to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 9°F, with a flow rate of not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided at the coolant inlet side of the fuel warmer unit.

3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

3.4.2.1 Starting system. For diesel engine driven vehicles, a 12 or 24-volt starting system, with 12-volt direct current (dc) lighting system, and not less than a 60 ampere alternator, shall be furnished. Engine starting equipment shall include an ether starting system or glow plug. When specified (see 6.2), if an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type. Measured shot type ether systems shall be key operated or manually operated from the driver's compartment, shall have a reservoir of not less than 12 fluid ounces, and shall be inoperative with the engine warm.

3.4.2.2 Ignition system. For gasoline engine driven vehicles, a 12-volt dc ignition system shall be furnished. An alternator of not less than 40 amperes rated capacity, and which provides not less than 16 amperes dc output at normal engine idle speed, shall be furnished.

3.4.2.3 Lighting. All vehicles lights, reflectors, and wiring shall be as specified herein and shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.19, 391.20 and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on vertical surface of rub rails (unless recessed and fully protected) or mounted on vehicle bumpers.

3.4.2.4 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. The vehicle shall be provided with double-faced front signal units and single-faced rear signal units installed in accordance with SAE J588, except that turn signals shall not be mounted on engine compartment hood. Turn signal operating units shall have visible and audible flash indicator.

3.4.2.5 Lighting cable. When specified (see 6.2), a lighting cable, conforming to SAE J1067, shall be furnished. The cable shall be coiled and shall be not less than 110 inches

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long when fully extended. Both ends of the cable shall be equipped with a round plug conforming to SAE J560. The plugs shall be equipped with a grip for withdrawing from the connector sockets. The cable shall be packaged and stowed in the vehicle tool compartment.

3.4.2.6 Batteries. Each battery shall be of 12 volt potential. The total reserve capacity rating and total cold cranking ampere rating at 0°F, both measured in accordance with SAE J537, shall be not less than specified in table III. Batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131.

TABLE III. Batteries.

Designation	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel	320	1200
Gasoline	100	450

* 3.4.2.7 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.4.3.1 Air cleaner. An air cleaner shall be furnished. When equipped with an oil-bath type, oil capacity shall be not less than 1 quart.

* 3.4.3.2 Fuel tank(s). Fuel tank(s) shall have not less than 43 gallon total capacity. When more than one tank is furnished on diesel engine driven vehicles, means shall be provided to assure equalized fuel level in both tanks. When more than one tank is furnished on gasoline engine driven vehicles, a selector valve connecting either tank to engine fuel intake shall be provided and means shall be provided to monitor the fuel level of either tank from a single fuel gage.

* 3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. Vertical mufflers, capable of being reached easily by personnel entering or leaving either aide of the cab, shall be provided with a heat shield. Vertical mufflers shall be provided with a hinged rain cap.

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* 3.4.5 Transmission. Unless otherwise specified (see 3.4.5.2), a manually shifted transmission shall be provided on the vehicle. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The vehicle shall be provided with not less than seven forward speeds. A multispeed transmission or auxiliary transmission shall be furnished. Gear ratios in transmission and axles shall be matched to provide a progressive shifting pattern throughout the complete range, and shall provide required vehicle performance (see 3.3.1). The transmission shall be provided with SAE J704 power takeoff openings. The transmission shall provide for maximum ease of shifting in all speeds.

3.4.5.1 Clutch. The clutch shall be the largest capacity clutch offered for the vehicle and engine furnished with torque capacity exceeding maximum delivered engine torque.

3.4.5.2 Automatic or semiautomatic transmission. When specified (see 6.2), the vehicle shall be provided with automatic or semiautomatic transmission. The transmission shall provide continuous drive. The transmission shall include a hydraulic torque convertor and not less than four forward gear ratios. Normal driving range selector position shall provide not less than four gear ratios without movement of the selector. A drive gear for the power takeoff shall be provided.

* 3.4.5.3 Power takeoff. The power takeoff shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver.

3.4.6 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.4.7 Frame. The vehicle shall be provided with a heavy duty main frame or frame reinforcements. Heavy duty frame shall have frame rai1s of greater section modulus than the manufacturer's standard for the vehicle furnished and shall provide structural strength at least equivalent to the reinforced frame specified herein. The reinforced frame shall have reinforcements extending at least from the rear of the front-suspension, rear hanger bracket to the front of the rear-spring, front hanger bracket. Reinforcements shall provide sufficient structural strength in the chassis frame through increased resisting bending moment to at least equal the loads imposed, with the vehicle loaded to provide specified GVW and in maximum high lift dumping angle. Frame rails shall not project beyond the rear end of the body.

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3.4.8 Suspension. The vehicle shall be equipped with a suspension system, with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the vehicle loaded to specified GVW. When suspension is rated at the spring pads, unsprung weight shall be deducted. Vehicle shall be equipped with hydraulic, double-acting shock absorbers at the front wheels. Auxiliary springs shall be furnished.

* 3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the vehicle loaded to specified GVW. Unless a two speed axle is specified, a single reduction rear axle shall be furnished.

* 3.4.9.1 Two speed axle. When specified (see 6.2), a two speed axle shall be furnished, equipped with electric vacuum, or air shift, and provided with ratios which will permit proper gear splitting. Gear ratio shall provide performance specified in 3.3.1.

3.4.9.2 Traction control. The vehicle shall be furnished with a traction control on the rear axle. The traction control shall actuate automatically to insure that power is transmitted to the wheels having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

3.4.10 Wheels, rims, tires, and tubes. The vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association recommendations for the type and size of tires furnished. Rim and disc wheel sizes shall be the same for all wheels on each vehicle. Tire size and load range (ply rating), shall be the same for all tires on the vehicle. When specified (see 6.2), the vehicle shall be equipped with wide base type tires and wheels for the front and rear axle in lieu of duals. Wheels shall be interchangeable without the use of an adapter. When specified (see 6.2), disc type wheels shall be furnished.

* 3.4.10.1 Tires. Steel belted radial or, when specified (see 6.2), bias ply tires shall be furnished. Tires shall be tube or tubeless type with highway trend. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the vehicle loaded to specified GVW. Tires shall conform to Tire and Rim Association recommendations or to ZZ-T-381 with a size designation system the same as the Tire and Rim Association.

* 3.4.10.2 Inner tubes. When tube type tires are furnished, inner tubes shall be of heavy duty type and shall be of proper size for tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

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* 3.4.10.3 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be furnished and stowed on the vehicle. When a spare tire assembly is specified (see 3.4.10.4), a spare wheel or rim shall be furnished. The spare wheel or rim shall be the same as mounted on the front and rear axles.

* 3.4.10.4 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare wheel or rim shall be the same size tread design and ply rating as the tires furnished on the vehicle.

3.4.10.5 Tire chain clearance. Tire chain clearance in accordance with SAE J683 shall be provided. Allowance for spring deflection shall be included.

3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43 and 393.45 through 393.52.

3.4.11.1 Service brakes. The vehicle shall be equipped with full-air brakes. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head-type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 7 1/4 cubic feet per minute (cfm)
(b) Air storage reservoir(s) with not less than 2,000 cubic inches total capacity, each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank
(c) Foot control, suspended or treadle type
(d) Air control valves
(e) Air pressure gage, visible to the driver
(f) Low-air-pressure warning, visible and audible
(g) Service brake stop lamp switch
(h) Alcohol aspirator, with unbreakable transparent container
(i) Automatic moisture ejector.
(j) When specified (see 6.2), in lieu of subparagraph (h), an air dryer shall be furnished. The air dryer shall be of the desiccant type.
* (k) When specified (see 6.2), automatic slack adjusters shall be furnished.

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3.4.11.2 Trailer brake control system. In addition to the components specified in 3.4.11.1, a trailer brake control system shall be furnished when a trailer towing package is required (see 3.1.1.9). The trailer brake control system shall include:

(a) Identification of emergency and service lines
(b) Coincident control of trailer brakes with prime mover foot control
(c) Independent hand control for trailer brakes
(d) Prime mover protection valve with dash control and automatic breakaway feature
(e) Trailer stoplight control operable with foot brake and with hand control for trailer brakes
* (f) Two SAE J844 coiled, air hoses, not less than 110 inches long when fully extended, with SAE J318 gladhand couplers on both ends of hoses. The hoses shall be packaged and stowed in the vehicle tool compartment for shipment
* (g) Air connectors for trailer with SAE J318 gladhand couplers, mounted at the rear of the vehicle, located to preclude interference with trailer
(h) Dummy gladhand couplers with chains
(i) Prime mover only, parking brake valve to permit prime mover parking brakes to be applied while charging the trailer air brake system.

3.4.11.3 Brake controls for use from a towing vehicle. When specified (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, relay emergency valve with no-bleed-back feature (except when spring applied emergency brake is furnished), additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not, compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

3.4.12 Cab. The manufacturer may furnish any standard full-width non-tilting model cab for the vehicle. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, with at least the curbside door equipped with external, key-operated lock. Drip rails shall be installed above the cab doors. Unless otherwise specified, the cab shall have upholstered, full-width, adjustable seat and back or individual, adjustable, driver's seat and individual passenger seat. When specified (see 6.2), an individual, adjustable driver's seat and an

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individual passenger seat shall be provided. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be furnished. Interior lighting shall be provided. Manufacturer's standard fastenings and two pairs of seat belts shall be installed on the vehicle. Safety grips, shall be provided on each side of the cab to assist personnel in entering and leaving the cab.

3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type and operated by either air or electric motor(s).

3.4.15 Bumper. Unless the bumper is an integral part of vehicle cab, a channel type front bumper shall be provided.

* 3.4.16 Tool stowage. Stowage space of sufficient size to accommodate vehicle jack, hand tools, antiskid chains, and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. Stowage space for these tools may be furnished inside the cab. When stowage space for these tools is located outside of cab, it shall be weatherproof and shall provide for locking with a padlock.

3.4.16.1 Tools. When specified (see 6.2), each vehicle shall be furnished with tools required for exchanging mounted tire assembly with, the spare assembly, and shall include at least a hydraulic jack, jack handle and wheelnut wrench. The jack shall be of such closed height as to permit its location under axle, or other satisfactory lift point at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of loaded vehicle to a height adequate to permit removal and replacement of wheel and tire assembly.

3.4.17 Heater and defroster. The vehicle shall be provided with a hot water heater with fresh air intakes and discharge outlets to floor and to windshield defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. All instruments shall be visible to driver when seated in driving position.

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3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
(d) Oil pressure gage or indicator
(e) Engine temperature gage or indicator
(f) Speedometer with recording odometer
(g) Dual sunvisors
(h) Driver's compartment ventilator other than window
(i) Tachometer (not required for gasoline engine driven vehicles).

* 3.4.20 Rearview mirrors. Outside rearview mirrors shall be mounted on each side of the cab. The mirrors shall be of the combination type having flat and convex areas enclosed in a common housing. The flat portion shall have not less than 50 square inches of reflective area. The convex portion shall have not less than 20 square inches of reflective area. The mirrors on conventional cabs shall have not less than three supporting arms.

3.4.21 Horn. The manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition, an air operated horn shall be furnished.

* 3.4.22 Engine hour meter. When specified (see 6.2), an engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

3.4.23 Backup alarm. The vehicle shall be provided with an audible, pulsating, signaling device (electrical or mechanical) to caution personnel when the vehicle is in reverse gear operation.

3.4.24 Leveling jacks. A hydraulic or mechanical stabilizing jack with pad shall be permanently mounted at each rear corner of the vehicle. The stabilizers shall be adjustable and shall be capable of swinging to a stowed position. Stabilizers in stowed position shall not protrude below the bottom of the chassis frame assembly. Mechanical type stabilizers shall be

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equipped with a removable crank handle. The vehicle shall be adequately reinforced at areas of jack installation to preclude damage to body and chassis assembly. The stabilizing jacks shall be of sufficient capacity to withstand the loads imposed during all operations of lifting and dumping without permanent deformation.

3.5 Body types.

* 3.5.1 Coal dump body (type I). The vehicle shall be equipped with a hydraulically operated coal dump body. The body shall have a total water level capacity of not less than 7 cubic yards (minus 5 percent). The body shall have two compartments and shall be equipped with a mechanically operated compartment divider. The vehicle shall have a cab-to-axle dimension of not less than 84 inches. Inside length shall be not less than 103 inches and inside width shall be not less than 84 inches. A rear bumper is not required.

3.5.1.1 Body construction. Body floor, sides, compartment divider, and front head shall be of not less than 8 gage (0.1644 inch) steel. The front head shall be capable of withstanding a horizontal static load equal to 1/2 the payload capacity of the vehicle without permanent distortion. When the body floor is constructed in two or more pieces, an automatic seam weld, having full penetration is acceptable without reinforcement. When seam is hand welded, underside of joint shall be reinforced with 3-inch wide, 8 gage (0.1644 inch) butt plates, welded between crossmembers. Full-length, channel formed, rub rail of minimum width to cover rear dual tire treads shall be provided. Not less than three triangular or box section side braces per side, of not less than 10 gage (0.1345 inch) steel, shall be equally spaced, between body front head and full box-type rear corner posts, and welded to side plates. Front head sheet shall be formed or reinforced for rigidity. Front head and tailgate shall be not less than 6 inches higher than the sides. Sides shall have pockets provided at each end for insertion of side boards. The interior of body shell and side reinforcements shall be welded with continuous welds. The rub rail shall be attached using continuous welds, or with intermittent welds for not less than 50 percent of the edge length of contact edges. The body shall have sloping running boards and sloping horizontal tailgate braces to minimize the buildup of dirt.

3.5.1.2 Tailgate. The tailgate panel shall be not less than 8 gage (0.1644 inch) steel. The tailgate shall be double acting, opening from top and bottom. The tailgate shall have heavy duty hardware, heavy duty support chains, and heavy duty tailgate latch operable by a control at the left front corner of the vehicle body. The tailgate shall be reinforced to prevent deformation under load. The tailgate shall be equipped with the manufacturer's standard built-in coal chute with a manually operated sliding door and a coal deflector. The coal chute shall not interfere with the normal use of the tailgate.

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3.5.1.3 Body understructure. Body longitudinal sills, each having a minimum section modulus equivalent to that provided by 7-inch, 9.8 pounds-per-foot I-beam, shall be provided to support hoist load. Not less than eight crossmembers, each having a minimum section modulus equivalent to that provided by a 4-inch, 5.4 pounds-per-foot channel, shall be provided to support an evenly distributed load of not less than 12,000 pounds plus the weight of the dump body. Crossmembers shall be welded to the body shell with not less than 4-inch lengths of weld, front and rear of both ends of each crossmember, and with staggered intermittent welds on not more than 12-inch centers. Contact edges of crossmembers with longitudinal sills and contact edges of welded reinforcements shall be welded for not less than 50 percent of the edge length.

3.5.1.4 Cab protector. A cab protector shall be attached to the front end of the body. The cab protector shall extend the full width of the cab and not less than 1/2 the distance from the back of the cab to the cab windshield. The cab protector shall be not less than 8 gage (0.1644 inch) steel. The cab protector shall be capable of supporting an evenly distributed load of not less than 1,500 pounds. The cab protector is not intended to be used for additional payload capacity.

* 3.5.2 Cargo dump (type II). The vehicle shall have a cab-to-axle dimension of not less than 84 inches. The load space shall be not less than 120 inches in length and 88 inches in width. Body side panels shall be not less than 24 inches in height. The body shall be of all-metal welded construction with members of minimum gage thickness as specified in table III for carbon steel; high tensile steel may be furnished in two gages lighter weight, in accordance with U.S. Standard gage sizes. Drain holes shall be provided in the front of the body floor. The top edge of body side panels shall be flanged and have five openings provided on each side for body bows having cross sectional dimensions of 1 inch by 2-1/2 inches. Lower bow sockets with bow stops shall be provided. Tiedown devices for securing a tarpaulin shall be provided. Structural sections of equal thickness may be used in lieu of framed sections specified in table III. The front panel shall be capable of withstanding a horizontal static load equal to 1/2, the payload capacity of the vehicle without permanent distortion.

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TABLE III. Framing gages.

Framing member	U.S. Standard gage no.	Equivalent inches
Floor	8	0.1644
Tailgate	8	0.1644
Crossmembers	10	0.1345
Side and end rails	12	0.1046
Longitudinal sills	10	0.1345
Reinforcements	10	0.1345
Posts	12	0.1046
Side and front panels	12	0.1046

* 3.5.2.1 Tailgate. The tailgate shall be hinged at the bottom and provided with duck covered stay-chains of a length to retain the tailgate in a horizontal position. The rear lights and reflectors shall be recessed in the rear bolster and shall be positioned to preclude damage by lowering the tailgate.

3.5.2.2 Body frame. The body shall be constructed with welded steel subframe. There shall be not less than 8 full-width crossmembers, including ends, and stub crossmembers as required to provide rigidity. The crossmembers shall be of full channel, pressed steel construction, reinforced by gusset plates or brackets at point of attachment to longitudinal sills. Crossmembers shall be welded to the body shell with not less than 4 inch lengths of weld, front and rear of both ends of each crossmember, and with staggered intermittent welds of not less than 4 inch lengths, on not more than 12 inch centers. Contact edges of the crossmembers with longitudinal sills and contact edges of welded reinforcements shall be welded for not less than 50 percent of the edge length.

* 3.5.3 Stake dump body (type III). Type III vehicles shall have rear-of-cab to rear-axle and body dimensions specified in table IV. Rear bumper shall be furnished. When specified (see 6.2), the center racks on both sides shall be the manufacturer's standard swing type, for easy side loading.

* TABLE IV. Type III dimensions.

Dimension	
Cab-axle dimension, minimum (inches)	84
Overall platform length, minimum (inches)	120
Overall width, minimum (inches)	92
Rack height, minimum (inches)	42

* 3.5.3.1 Body. The body shall be provided with steel subframe, wood or steel platform, and side and end racks.

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* 3.5.3.2 Body frame. Body framing shall be a completely welded structure with members of minimum gage thickness specified in table V for carbon steel; high tensile steel may be furnished in two gages lighter weight in accordance with US Standard gage sizes. There shall be not less than a full width crossmembers, including ends and stub crossmembers as required for proper spacing over axle. Crossmembers shall be of the full channel construction, reinforced by gusset plates, or brackets at points of attachment to longitudinal sills, and contact edges of welded reinforcements shall be welded for not less than 50 percent of the edge-length. Longitudinal sills shall be constructed of structural steel channel or formed channels. Formed channel sills shall be reinforced within the sill, at each crossmember or body mounting point, with formed channel reinforcements.

TABLE V. Framing gages.

Framing member	US Standard gage number	Equivalent inches
Crossmembers	10	0.1345
Side and end rails	10	0.1345
Longitudinal sills	8	0.1644
Reinforcements	8	0.1644
Rack posts	11	0.1196
Rack posts with reinforced lower section	12	0.1046

* 3.5.3.3 Flooring. Unless a composite type of construction is specified (see 3.5.3.6), the platform shall be floored with wood or steel, at the manufacturer's option. Wood parts shall be treated in accordance with MIL-STD-1223.

* 3.5.3.4 Wood floors. Wood platforms shall be floored longitudinally with either ship-lap or tongue-and-groove joints. Wood flooring shall be of hardwood or dense southern yellow pine not less than 1-5/16 inches thick (finished dimension).

* 3.5.3.5 Steel floors. Steel floors shall be 1/8 inch thick, one or two piece diamond tread with additional lateral support provided at the wheel wells. Two piece floors shall be spliced longitudinally and completedly welded the full length of the splice.

3.5.3.6 Composite floors. When specified (see 6.2), composite type of floor construction shall be furnished. Composite type shall include not less than eight steel flanged (hat shaped) channels extending the full length of the platform. Channels shall be constructed of not less than 14 gage (0.0747 inch) steel, 1-1/2 inches wide at the top, of a height flush with or slightly below

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the floor boards, and with bottom flanges not less than 7/8 inch wide. Floorboards shall be accurately fitted between the hat shaped channels with 1/4 inch minimum diameter, self-tapping, corrosion resistant screws, two at each cross sill and on each side of each butt joint. Screw heads shall be countersunk.

* 3.5.3.7 Side and end racks. A full width front rack section, not less than three removable rack sections on each side, and two removable rack sections across the rear, shall be provided. Each rack shall be equipped with a locking device to lock the rack to the body. Body hardware shall be attached to the rack slats with not less than 3 inches total weld for each fastener. Upright posts shall be steel sections. Rack slats shall be of steel sections not less than 18 gage (0.0478 inch) thick with not less than 3 reinforcing ribs. Width of individual slats shall be manufacture's standard, providing that total of slat widths is not less than 60 percent of total rack height. Not less than four slats per rack shall be provided. Slat edges and end corners shall be rounded or enclosed to protect cargo and personnel from sharp edges. Slats shall be riveted, bolted or welded to the inside (loadside) of the upright posts, with rivet or bolt heads against slat. When welded construction is used, not less than 4 welds shall be applied at each up-right post and slat intersection. The front rack section shall be capable of withstanding a horizontal static load equal to 1/2 the payload capacity of the vehicle without permanent distortion of the rack section or its mountings. The rear end racks shall be the manufacturer's standard swing type, hinged to each side rack. Locking devices shall be provided near the center of the rear racks to lock the racks closed and to lock the racks to the body. All locking devices shall be operable from the ground.

* 3.5.4 Hydraulic hoist. The body shall be provided with a hoisting mechanism capable of lifting the fully loaded body to a level height of 10 feet, measured from the ground to the floor of the body, and dumping the load. The hoist shall be the manufacturer's standard with a minimum lifting capacity rating of class 40 in accordance with the TBEA Hoist Classification Chart. The hoist shall lift the body through a minimum dumping angle of 45 degrees. The power takeoff, pump and valve shall be the manufacturer's standard for the hoist model furnished. Sway bracing within the hoist structure shall be provided. Means shall be provided to couple the truck body to the chassis frame automatically when in the retracted position. A positive mechanical safety device shall be furnished to insure that the power takeoff is disengaged before the vehicle can be put in motion and that the vehicle cannot be put in motion with the power takeoff engaged. The vehicle shall be equipped with positive stops to retain the lift and dump

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mechanism within the extremes of its required travel. Ground clearance of the hoist mechanism components shall be not less than 9-1/2 inches.

3.5.4.1 Controls. Raising, lowering, and stopping the hoist mechanism shall be regulated by one set of controls located inside the cab and shall be readily accessible to the operator when seated in the driving position. The controls shall, upon release of applied force by the operator, stop and hold the truck body at any point within the lifting, dumping, and lowering ranges. The controls shall be of the deadman type requiring continuous application of pressure.

* 3.5.4.2 Operating speeds. The lifting and lowering speeds shall be variable between 0 and maximum of 10 feet per minute.

3.5.4.3 Normal slip. With hydraulic pressure applied, the rate of measured hydraulic pistons shall not exceed 1/2 inch in 2 hours when the dump body is in the maximum dump angle and loaded with rated payload.

3.5.4.4 Safety lock. A mechanical safety lock, permanently affixed to the body, shall be furnished. The safety lock shall provide positive retention of the body in the up position for servicing or repair. The safety lock mechanism shall not interfere with the operation of the body under any operating conditions.

3.5.5 Hydraulic system.

* 3.5.5.1 Hydraulic pump. A hydraulic pump with compatible pressure and flow capacities to operate all hydraulic components and accessories shall be furnished. The hydraulic pump shall be driven by a heavy duty power takeoff unit attached to the truck-chassis transmission. Split-shaft power takeoff is not acceptable. Chain drives are not acceptable. The hoist pump shall be of the gear-type design. The pump pressure continuous rating shall be not less than that recommended by the lift platform manufacturer, with an intermittent rating of 500 pounds per square inch (psi) over the continuous rating. The system hydraulic pressure shall be limited to not more than 2500 psi under any operating conditions. The pump system shall include a built-in adjustable pressure relief valve(s) or unloading valve(s), with the ranges of adjustment which allow the loads specified herein. The hydraulic pump shall be capable of operating in ambient temperatures from minus 40°F to plus 125°F.

* 3.5.5.2 Automatic throttle actuator. An automatic hydraulic or electric throttle actuator shall be furnished. The actuator shall increase automatically the truck engine speed when the pump is being loaded, and automatically decrease the speed of the engine when no load is being imposed.

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* 3.5.5.3 Hydraulic reservoir. A hydraulic reservoir shall be furnished. The reservoir shall be equipped with a drain, a filler opening cap attached to the reservoir by a corrosion resistant chain, and internal baffles to prevent excessive surging of the fluid. A dip stick or sight gage shall be provided to observe the level of fluid when the lift body is in its lowered position. A tank vent shall be furnished and installed to preclude weather and foreign matter entering the reservoir. Provisions shall be made for inspecting and cleaning the reservoir without loss of fluid in the system. The reservoir shall be provided with a photo-etched, corrosion resistant plate to indicate the type of hydraulic fluid. The reservoir shall be protected to preclude damage.

* 3.5.5.3.1 Hydraulic fluid. Unless otherwise specified, hydraulic fluid conforming to the manufacturer's standard or recommended requirements shall be provided in the hydraulic system. When specified (see 6.2), hydraulic fluid conforming to MIL-H-46170 shall be provided in the hydraulic system.

* 3.5.5.3.2 Oil filter and strainer. A full flow type 10-micron discharge oil filter with a capacity of not less than three times the pump flow shall be provided in the return life to the reservoir. A 100 mesh suction line strainer with a capacity of not less than two times the pump flow shall be provided in or near the hydraulic reservoir. The return line oil filter shall be equipped with a condition indicator gage and a 15 psi internal bypass. The gage shall be so located as to be easily readable with the body in the down position. Filter and suction strainer shall be accessible for cleaning or replacement of the filter cartridge.

* 3.5.5. 4 Hydraulic fluid temperature. The hydraulic fluid temperature shall be maintained within safe limits to preclude seal damage and faulty operation of hydraulic valves and controls. Excessive heating of more than 55° above ambient shall not be acceptable under any operating condition.

3.5.5.5 Hydraulic lines and fittings. Hydraulic lines shall be located to facilitate maximum ease of removal and installation, or visual inspection for failures and maximum protection from damage during operation of the body assembly. Flexible, hydraulic hose or collector rings shall be used where flexing of hydraulic lines is encountered and where installation of components requires a flexible coupling. Standard straight-thread, flared type

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fittings or adaptions to flared type fittings, or tapered pipe thread with dry seal fits shall be furnished. Hydraulic lines and fittings shall be air and liquid tight and shall be installed to preclude vibration and chafing. High pressure hoses which are located in areas where chafing may occur shall be covered with spiral type hose guards. The system shall be designed to preclude hydraulic transients. Provisions shall be made for bleeding air from the hydraulic system and completely draining all hydraulic lines, when necessary. A manual type bleeder shall also be provided as on-vehicle equipment on closed-centered systems. Hose assemblies and all seamless steel hydraulic lines shall be adequate to provide the rate of flow, temperature, and pressures required to operate the system smoothly and efficiently. Galvanized pipe and copper tubing shall not be acceptable. A pressure gage shall be provided on the discharge side of the hydraulic pump. Accumulators shall not be furnished. The system shall be designed to a safety factor of 3 for normal operating conditions.

* 3.5.5.5.1 Hydraulic valving. An in-line relief valve shall be provided, fully adjustable, and compatible with the working loads and test loads specified herein. The valve shall have provisions for a safety lock wire on the adjustment portion of the valve to permit sealing and locking at the proper working pressure after test. A normally closed, 12-volt dc, solenoid operated check valve or valves shall be provided to lock the cylinder(s) in the event of a failure in any other portion of the hydraulic system. If the check valve is not integral with the cylinder, a minimum number of fittings shall be used between the check valve and the cylinder, with a burst strength of at least three times the maximum operating pressure. A normally open, 12-volt dc, solenoid operated, two position, two way valve with pilot assist, shall be provided to raise and lower the body. The high pressure hydraulic system shall be provided with a gage port to accommodate a 2-1/2 inch face, 0-3000 psi, glycerin filled, stem-mounted gage. The gage shall be easily visible with the body in the down position. The gage need not be furnished.

3.5.5.6 Hydraulic cylinders. The lift mechanism shall be actuated by single or dual cylinders. When dual cylinders are used, they shall be so connected to a hydraulic manifold to assure equal pressure and simultaneous operation of both cylinders. Cylinders may be single, multiple telescoping, or other special design. All seals, rings, gaskets, and other related components shall be designed for easy removal and replacement. Metal nonflexing hydraulic lines shall be used where flexing is not required during operation. An armored or otherwise protected flexible hose shall be used where flexing is required. Provisions shall be made for bleeding the system at all necessary points without removing or loosening any component of the system. The cylinder(s) shall be positioned to preclude the cylinder(s) extending above the body

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floor. The hoist cylinder(s) shall be furnished with an automatic cushioning of stroke approaching the fully extended and retracted positions of the body. The hydraulic system shall be designed to preclude hydraulic transients. Automatic pilot check valves shall be provided to preclude body descent in the event of a hydraulic pressure loss. A manual device or valve shall be incorporated into the system to lower the body in case of hydraulic pressure loss. The device will bleed the fluid back into the reservoir. The device shall provide metering characteristics.

3.5.5.6.1 Hydraulic cylinder valving and manifold. A check valve, a return line with filter having antiflow-back characteristics or a ball type shut-off valve shall be provided between the reservoir and the return line filter to enable shut off of the oil supply to facilitate changing the element in the filter. A fixed orifice device or an adjustable throttle valve (flow control) shall be located as close as practicable to the lift cylinder(s) to limit and control the downspeed. The throttle valve shall be rated for not less than three times the working pressure and shall be equipped with a lock screw or jam nut to hold the desired preset. When two lift cylinders are provided, the manifold shall distribute equal pressure and flow rate and furnish simultaneous operation of the cylinders.

* 3.6 Body mounting. The body shall be mounted to the hoist unit in accordance with the hoist manufacturer's recommendations and shall be reinforced, when necessary, for added strength on hoist operations. Rear body mounting shall include hinges securely welded to the body longitudinal sills, a connecting cross shaft, and a plate securely bolted to the chassis main frame rails.

3.7 Welding. Adequate preparation of metal prior to welding of the body subframe, hoist, elevating body, and components shall be provided. The surface of parts to be welded shall be free of rust, scale, paint, grease, and other foreign matter. Welding rods shall be compatible with the metal being welded. All critical arc welds shall be magnaflux or dye penetrant inspected prior to painting. The following defects shall not be acceptable:

(a) Welds which are ground flush with metal
(b) Slug or plug welding
(c) Undersize of fillet welds, overlaps, inadequate fusion and penetration, undercutting of weldments and vertical down welding
(d) Unremoved slag and spatter
(e) Cracks
(f) Spot, tack, or intermittent welds for strength.

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Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Intermittent welding shall be permissible when in accordance with 3.6.1 and 3.6.2.

3.7.1 Welding procedures. The welding procedure shall be recorded prior to production of any weldment, but no procedure qualification other than workmanship specimens shall be required.

3.7.2 Factors. The factors listed below, which are applicable, shall be included in the recorded welding procedure:

(a) Joint sketch
(b) Joint type and preparation
(c) Base metal type
(d) Filler metal, size and type
(e) Position of welding
(f) Arc-voltage, current voltage, and polarity
(g) Manual or machine
(h) Travel speed, machine only
(i) Pre-heat and post heat.

3.7.3 Workmanship specimens. Workmanship specimens shall be prepared by the manufacturer for the body, hoist, subframe, and components to simulate all welding conditions. The specimens shall represent the minimum acceptable weld quality and cleaning procedure to be used in production. Upon submittal of these specimens by the contractor, the quality level and identification shall be subject to approval by the Government. The specimens shall be prepared using the welding procedures to be employed in production. Specimens prepared to represent multiple pass, welds shall be made in such a manner as to leave at least two inches of each layer of weld metal exposed. Approval specimens shall be retained at the contractor's plant.

3.8 Stress analysis. A stress analysis for the body shall be submitted to the engineering office of the procuring activity. The stress analysis shall include the weight of all major items permanently installed on or within the vehicle, and shall show the location of the center of gravity of such components relative to the vehicle axles (raised and lowered positions). In addition, the data shall include the value of stresses in the vehicle chassis frame and its subframe, and at the trunnions of the cylinders in the lowest position and when being raised to its highest position. The analysis shall verify that the yield point of the material used is not exceeded under the conditions specified herein. Stress analysis shall be submitted with the vehicle questionnaire.

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3.9 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle for operational use including at least the following: alining of lights; adjustment of the engine, electrical and brake system; filling and charging of battery; alinement of front wheels; inflation of all tires; complete lubrication of chassis, engine, and running gear, with lubricants recommended for the ambient air temperature at the delivery point; servicing of cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of windshield washer reservoir with water and appropriate additives.

3.10 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to, determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First vehicle preliminary inspection.

* 4.3.1 Welding procedure and workmanship specimen evaluation. Prior to start of fabrication of the subframe, hoist, body, components and assemblies, an inspection shall be conducted by the Government to evaluate workmanship specimens and welding procedures.

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* 4.3.1.1 Macro-etching. Workmanship specimens shall be cross-sectional and macro-etched. Both parts of the specimens shall be made available to the Government representative for examination.

4.3.1.2 Radio frequency suppression verification. The vehicle manufacturer shall furnish with the questionnaire (commercial vehicle engineering data) a description of all interference suppression devices used to suppress the vehicle electromagnetic radiation in accordance with SAE J551. Data shall be sufficient for visual determination that all required suppression devices are installed on the vehicle.

* 4.3.2 In-process inspection of body and components. The first complete assembled body and components (sub-frame, hoist, body, components and assemblies) shall receive an in-process inspection by the Government prior to prime painting. The inspection shall be conducted to evaluate workmanship and to determine conformance to contractual requirements.

* 4.3.2.1 Structural inspection. Dye penetrant or magnaflux shall be applied on the critical areas of welds prior to painting. A water soluble penetrant shall be applied on all critical areas. The water soluble penetrant or magnaflux shall provide a means to inspect for crack indications under dynamic and static conditions.

4.4 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformity to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

* 4.4.1 Weight and payload capacity. The first production vehicle shall be weighed to determine curb weight and distribution of curb weight on front and rear axles. The imposed loading on front and rear axle(s) will be computed to ascertain that the suspension, axles, and tires furnished are of adequate capacity to set contractual requirements. Distribution of weight will be computed by using the curb weight, the operator weight at 175 pounds, and a payload to provide the specified GVW.

4.4.2 Road test prior to operational tests. The first production vehicle shall be examined and road tested by the contractor to assure that the vehicle will operate in accordance with contract requirements. The vehicle, loaded with payload and with all equipment specified herein, shall be driven at speeds up to at least 50 mph, for a minimum distance of 50 miles. The

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distance traveled shall consist of highway, rough dirt roads, partly surfaced gravel roads, puddles, and sharp turns. For each 10 miles of travel at least one sudden stop shall be accomplished. Upon completion of the road test all equipment shall be thoroughly examined. The equipment shall show no evidence of damage; all body and cab doors shall properly open and shut; there shall be no evidence of mud or dust inside vehicles; and there shall be no evidence of racking of the body.

4.4.3 Operational testing.

* 4.4.3.1 Upsetting and range of movement. With the vehicle resting on approximately level ground and not fastened to any artificial base, the body shall be loaded with twice the payload capacity (GVW minus curb weight). The body shall be extended to its full height and dump positions, lowered to travel position and raised to increments of one third, two thirds, and to its full height and dump position and lowered to travel position.

4.4.3.2 Wind load test. Wind resistance shall be demonstrated with and without the payload. With the payload evenly distributed over the body load space, and the body elevated to its maximum specified height, measured from the ground, and the vehicle positioned whereby the greatest area is parallel to, the centerline of the vehicle, the specified wind load, generated by artificial means or by an, equivalent simulated wind, shall be applied to the vehicle from the sides (see 3.3.3.7). The contractor shall furnish to the engineering office of the procuring activity, calculations or a certified test report.

* 4.4.3.3 Elevating and dumping body speed test. With the vehicle resting on approximately level ground and the payload uniformly distributed over the body load space, the body shall be elevated to its maximum specified height and held for 2 minutes, plus or minus 1/2 minute, placed in dump position and then lowered to rest position forty cycles, with a 30 minute interval between each 10 cycles. Each cycle equals one complete elevation, dumping and retraction of the body. The test shall verify the requirements of' 3.3.3.4. The test shall be performed in an ambient temperature of 75°F plus or minus 10°F.

4.4.3.4 Creep test. With the vehicle resting on approximately level ground, and an evenly distributed payload placed on the body, the body shall be extended to its full height and held for not less than two hours. Vertical settling shall be not more than 1/8 inch in two hours (see 3.3.3.5).

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* 4.4.3.5 Dynamic conditions. With one and one-half times the payload on the body and with the body fully extended and operating through various positions, as deemed necessary by Government inspector, and with the body being raised, lowered and dumped for at least 1/2 hour, the water soluble penetrant and magnafluxed areas shall be inspected for any evidence of cracks. Any evidence of cracks or deformation shall be cause for rejection of the vehicle.

* 4.4.3.6 Static conditions. With the body in its most vertically extended position and with three times the payload placed on the body, water soluble penetrant and magnafluxed areas shall be inspected for evidence of cracks. Evidence of cracks or deformation shall be cause for rejection of the vehicle. At the manufacturer's option, dye penetrant and magnaflux inspection may be conducted prior to the first production vehicle inspection, provided the tests specified in 4.4.3.5 and 4.4.3.6 are conducted prior to the inspection; and records of the tests and the dye penetrant and magnaflux inspection witnessed by the Government are presented at the time of the first production vehicle inspection.

* 4.4.3.6.1 Critical areas. At least the following critical areas shall be inspected by dye penetrant or magnaflux:

(a) Sub-frame, trunnion, yoke, body cylinder guides, and attaching areas
(b) Struts, hoisting mechanism
(c) Structural steel of body and sub-frame
(d) Tailgate support areas, type I
(e) Platform support areas, types II and III.

* 4.4.3.7 Hoist stability. The hoist shall provide sufficient structural strength to preclude shifting laterally or fore and aft, without evidence of undue lateral flexing of the structure and without evidence of wear due to binding of the hoist members.

* 4.4.3.8 Controls. Controls shall be operated and shall meet or exceed the requirements specified herein.

* 4.4.3.9 Hydraulic system verification. The hydraulic system shall be verified to assure that all contractual requirements are met.

* 4.4.3.10 Body electrical system. The body electrical system shall be verified to assure that the system has sufficient capacity to meet the requirements specified herein.

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* 4.4.4 Servicing and adjustment of the hydraulic system. Servicing and adjustment of the hydraulic system shall be demonstrated as specified herein.

* 4.4.5 Final road test. The final road test shall be performed after 4.4.3.5 through 4.4.3.10 have been completed. The road test shall be as specified in 4.4.3, except the minimum distance shall be 20 miles.

* 4.4.5.1 Final operational test. The final operational test shall be performed subsequent to 4.4.5. The body shall be fully elevated, dumped and retracted with a uniformly distributed payload for not less than 30 cycles, with a 30-mintute interval between each 10 cycles.

* 4.4.6 Performance test. The body assembly shall be subject to, but not limited to, the following tests:

(a) Controls activate when the transmission is in neutral
(b) Controls return to neutral position when the operator releases the control
(c) Operate equipment using each set of controls separately
(d) Check to insure that the entire system can be operated smoothly and without jerking motions in any position
(e) Determine whether the body remains level in all positions with one and one-half the specified loads in the body.
(f) Check to determine whether the body can be safely lowered, from maximum reach position, with rated payload, with the hydraulic pump and engine not operating
(g) Check the hydraulic system, including the pump at 10 percent above the pump manufacturer's maximum operating pressure setting for normal workload conditions.

* 4.4.7 Body treatment and painting. The certification regarding body cleaning, treating, and prime painting, as required by MIL-STD-1223, shall be presented to Government representatives for examination and approval.

4.5 Production sample. Upon acceptance of the first production vehicle, the vehicle shall remain at the manufacturing facility as a production sample and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.6 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until

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corrective action has been taken. In addition, subsequent vehicles that do not conform to the accepted production sample vehicles shall be rejected until corrective action has been taken.

4.7 Inspection of production vehicles. The contractor's inspection system shall as a minimum assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required in 3.9 has been accomplished.

5. PACKAGING

* 5.1 Vehicle processing. The vehicle shall be processed for shipment from the manufacturer's plant to the initial receiving activity in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general nontactical use by the Government in transporting and dumping coal into bins, and cargo onto platforms and areas which are 10 feet above ground level height.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of vehicle required (see 1.2).
(c) Identification of appropriate service for painting and marking (see 3.1.1.1).
(d) Concealed marking, if required (see 3.1.1.1).
(e) Rustproofing, if required (see 3.1.1.2).
(f) Tropical rustproofing, if required (see 3.1.1.2).
(g) Heavy duty cooling system, if required (see 3.1.1.5).
(h) A trailer towing package, if required (see 3.1.1.9).
(i) Gasoline engine, if required (see 3.4.1.2).
(j) Silicone rubber hoses, if required (see 3.4.1.6).
(k) Power plant heaters, if required (see 3.4.1.7).
(l) If a battery heater is not required (see 3.4.1.7).
(m) A measured shot type ether system, if required (see 3.4.2.1).
(n) Lighting cable, if required (see 3.4.2.5).
(o) Automatic or semiautomatic transmission, if required (see 3.4.5.2).
(p) Two speed axle, if required (see 3.4.9.1).

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(q) Wide base type tires and wheels, if required (see 3.4.10).
(r) Disc type wheels, if required (see 3.4.10).
(s) Bias ply tires, if required (see 3.4.10.1).
(t) Spare wheel or rim, if required (see 3.4.10.3).
(u) Spare tire, if required (see 3.4.10.4).
(v) Desiccant type air dryer, if required (see 3.4.11.1(j)).
(w) Automatic slack adjusters, if required (see 3.4.11.1(k)).
(x) Brake controls for use from a towing vehicle, if required (see 3.4.11.3).
(y) Individual driver and passenger seats, if required (see 3.4.12).
(z) Tools, if required (see 3.4.16.1).
(aa) Air operated horn, in addition, if required (see 3.4.21).
(ab) Engine hour meter, if required (see 3.4.22).
(ac) Swing center racks, if required (see 3.5.3).
(ad) Composite type floor construction, if required (see 3.5.3.6).
(ae) Hydraulic fluid conforming, to MIL-H-46170, if required (see 3.5.5.4.1).

6.3 Performance prediction. Completed copies of SAE Truck Ability Prediction Procedure form and computation for tow speed requirements for each vehicle model furnished under contract shall be submitted as specified in the contract. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry air. The SAE Work Sheet Item I should include vehicle model number, wheelbase, engine model number, transmission model number, and vehicle type. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1	- Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	- Altitude Factor. 1.00
Table 3	- Rolling Factor. 1.613
Table 4	- Area Factor. 0.173
Table 5	- Velocity Factor. 250.0
Table 6	- Altitude Factor. 1.00
Table 7	- Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher 1000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 50 mph geared speed. For

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GVW and engine speed beyond the range of this table, factors shall be extrapolated.

Table 8 - Grade Factor. 0.75
Table 8A - Correction Factor. Not required.
Table 9 - Road Factor. 0.0.

* 6.4 Changes from previous issue. The margins of this document are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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